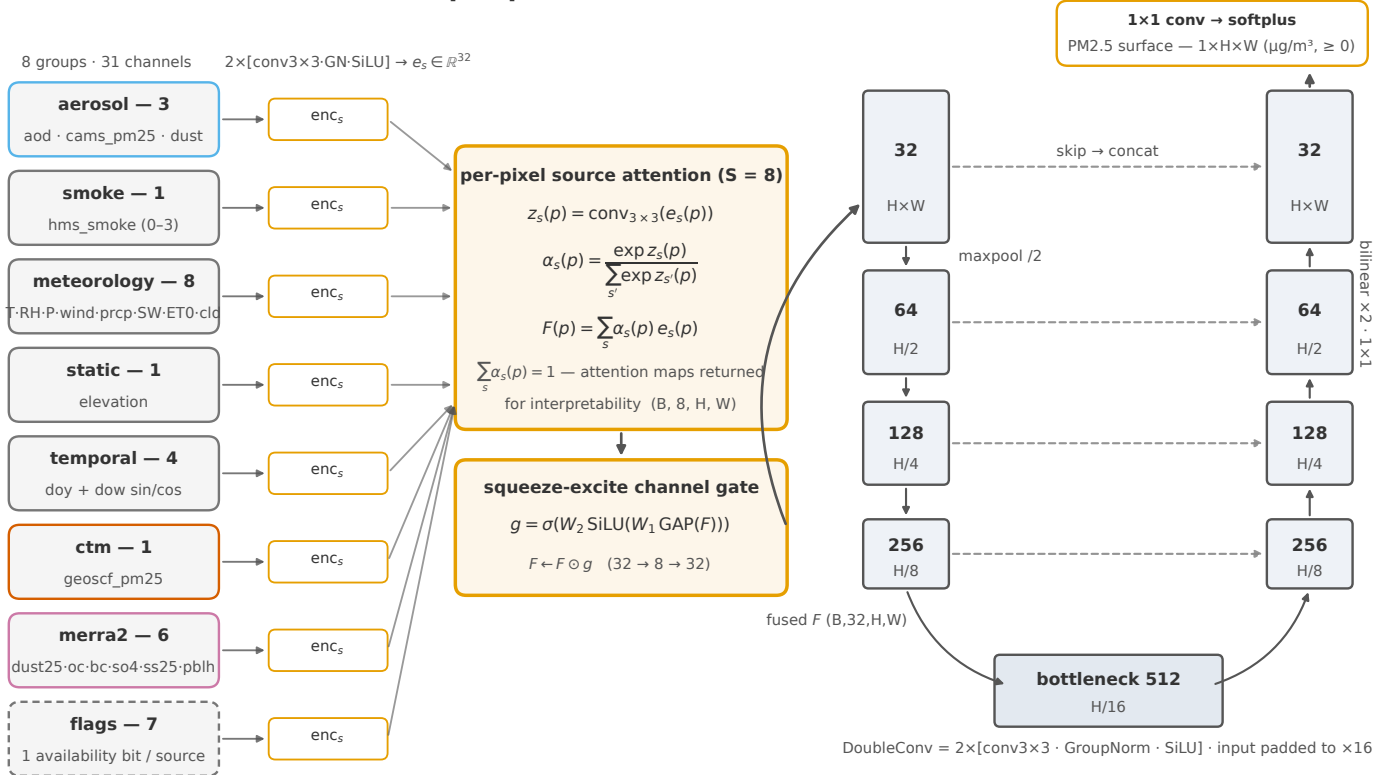


Tier 2 — FusionUNet: fusion → per-pixel attention → U-Net → PM2.5 surface



training: masked Huber ($\delta = 15$) at sensor pixels only · AdamW (lr $1e-3$, wd $1e-3$) · 5-epoch warmup → cosine · AMP · 96 px random crops

modality dropout: each source group zeroed w.p. 0.15 per batch after normalization — flags exempt, so the availability signal survives

flags group: 1 binary channel per source group, 1 when the source reported that day pre-fill (flag-and-fill)